

Mustafa Mudasser

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer (Data Platform) with 4+ years of Python production experience. Versatile engineer who solves problems across data pipelines, REST APIs, databases and cloud infrastructure. Strong background in platform engineering, data analysis and applying AI to turn raw data into faster insight.

Industry placement at JPMorgan Chase building a production Django platform with CI/CD and full TDD. MSc research applying AI and NumPy-based time-series analysis to 58-feature metadata processing. Works cross-functionally, communicates clearly, and takes genuine ownership of how platforms grow and scale.

TECHNICAL SKILLS

Languages: Python (primary), SQL, Bash

Frameworks: Django, FastAPI, Flask, Django REST Framework

Data and AI: NumPy, pandas, scikit-learn, SHAP, LIME, Hugging Face, time-series data analysis, metadata processing

Databases: PostgreSQL, MySQL, SQLite, relational schema design

Cloud and Infrastructure: AWS (EC2, S3, Lambda, RDS), cloud infrastructure management, on-prem Linux deployments, networking basics

DevOps: Docker, Jenkins CI/CD, GitLab CI (conceptual), Git, Linux, platform engineering

Testing: pytest, unittest, TDD, BDD, integration testing, peer code review

Other: REST API design, microservices architecture, data pipelines, Agile/Scrum, cross-functional collaboration

PROFESSIONAL EXPERIENCE

Software Engineer JPMorgan Chase London, UK Jan 2026 - May 2026

Industry Placement

- Built a production Django REST banking platform with PostgreSQL, implementing security-focused authorisation and time-gated API endpoints in a regulated, fully auditable environment
- Designed a 5-stage Jenkins CI/CD pipeline automating build, test and deployment; containerised the full platform engineering stack with Docker on Linux infrastructure
- Applied TDD and BDD across all features, writing unit and integration tests covering data validation, failure modes and edge cases in financial transaction flows
- Worked cross-functionally with product, engineering and business stakeholders to define API contracts, solve problems collaboratively and communicate complex technical decisions clearly

Python Engineer Grand Digitals Remote Jan 2025 - Dec 2025

- Designed Python microservices and REST APIs for high-volume transactional data pipelines on AWS cloud infrastructure, reducing manual processing by 40%
- Delivered 15+ platform solutions end-to-end — owning design, development, testing and deployment — in a versatile generalist engineer role across the full stack
- Built data analysis pipelines processing structured and time-series metadata from multiple sources, with error detection logic to flag data quality issues early
- Collaborated cross-functionally across distributed teams to drive code quality, documentation standards and reliability improvements

Python Developer Visnext Software Solutions Lahore, Pakistan Jan 2023 - Nov 2024

- Built FastAPI and Django microservices for financial clients, with PostgreSQL and MySQL databases and complex SQL queries across 30+ fields for analytical reporting
- Designed and deployed automated data analysis pipelines on AWS Linux infrastructure, reducing operational overhead by 90% through platform engineering and workflow automation
- Led peer code reviews enforcing unit testing standards, clean code practices and dependency management — contributing to team-wide engineering best practices

- Worked directly with stakeholders to gather requirements, solve data problems, and verify pipeline outputs against business expectations

Associate Software Engineer Airlift Technologies Lahore, Pakistan Jul 2021 - Jul 2022

- Built automated Python pipelines processing high-volume operational data and metadata in real time, developing core skills in data analysis and platform engineering
- Delivered reliable data transformation and reporting automation, improving consistency and reducing manual effort across the team

FEATURED PROJECT

Interpretable AI Trading Agent MSc Final Project, Bournemouth University 2025 - 2026

- Engineered 58 time-series features and metadata signals from raw cryptocurrency market data using NumPy, designed for computational efficiency and numerical precision in a high-volume data platform
- Applied AI (deep reinforcement learning with Hugging Face integration) to automate data analysis and generate insight from raw time-series data — directly paralleling how AI can extract more value from sensor and telemetry data
- Built SHAP/LIME explainability pipeline achieving 0.96 Pearson fidelity ($p < 0.0001$) across out-of-sample test periods — verifiable, auditable AI reasoning
- Full platform engineered end-to-end in Python: data ingestion, feature engineering, model training, inference and validation; version-controlled on GitHub with structured documentation

EDUCATION

MSc Data Science and Artificial Intelligence Bournemouth University Bournemouth, UK Jan 2025 - Jun 2026

Graduating June 2026. Modules: machine learning, data engineering, statistical modelling, applied AI, cloud computing.

BSc Computer Science University of Central Punjab Pakistan 2018 - 2022

Focus on software engineering, algorithms, networking, database systems and distributed computing.